3.7 Properties of Triangle

One important use of trigonometry is to solve practical problems that can be modeled by a triangle. Determination of all the sides and angles of a triangle is referred as solving the triangle. In any triangle, the three sides and three angles are called basic elements of a triangle. Pythagorean theorem plays a vital role in finding solution of the right triangle. The law of sines and the law of cosines are important tools that can be used effectively in solving an oblique triangle (a triangle with no right angle). In this section, we shall discuss the relationship between the sides and angles of a triangle and derive the law of sines and the law of cosines.

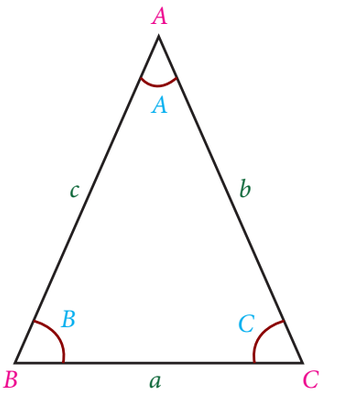


Figure 3.17

Notation: Let ABC be a triangle. The angles of $\triangle ABC$ corresponding to the vertices A, B, C are denoted by A, B, C themselves. The sides opposite to the angles A, B, C are denoted by a, b, c respectively. Also we use the symbol $\triangle$ to denote the area of a triangle.

The circle which passes through the three vertices of a triangle is called circumcircle of the triangle. The centre and the radius R of the circumcircle are known as circumcentre and circumradius respectively.

In $\triangle ABC$, we have $A + B + C = \pi$ and $a + b > c, \quad b + c > a, \quad a + c > b$.

The Law of Sines or Sine Formula

3.7.1 Law of Sines

The Law of Sines is a relationship between the angles and the sides of a triangle. While solving a triangle, the law of sines can be effectively used in the following situations:

(i) To find an angle if two sides and one angle which is not included, by them are given.
(ii) To find a side, if two angles and one side which is opposite to one of given angles, are given.

Theorem 3.1 (Law of Sines)**:** In any triangle, the lengths of the sides are proportional to the sines of the opposite angles. That is, in $\triangle ABC$, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$ where R is the circumradius of the triangle.

Proof. The angle A of the $\triangle ABC$ is either acute or right or obtuse. Let O be the centre of the circumcircle of $\triangle ABC$ and R, its radius.

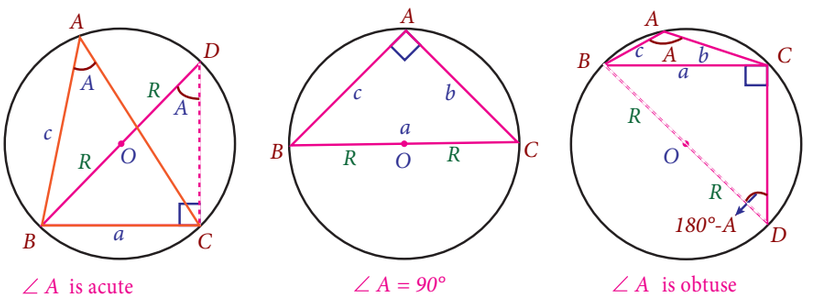


Figure 3.18